Experiment # 08

Configuring Static Routes Using Static Routing Protocol

Objective:

This Lab explains how to configure static routes.

Software Tools:

- Cisco Packet Tracer

Theory:

Static Routing Configuration

Static routing is the most secure way of routing. It reduces overhead from network resources. In this type of routing, we manually add routes to the routing table. It is useful where the number of route are limited. Like other routing methods static routing also has its pros and cons.

Routing Types:

- **Static routing**
  Static routing allows routing tables in specific routers to be set up by the network administrator.

- **Dynamic routing**
  Dynamic routing use Routing Protocols that dynamically discover network destinations and how to get to them. Dynamic routing allows routing tables in routers to change if a router on the route goes down or if a new network is added.

In Dynamic Routing, Routing Protocols running in Routers continuously exchange network status updates between each other as broadcast or multicast. With the help of routing updates messages sent by the Routing Protocols, routers can continuously update the routing table whenever a network topology change happens.

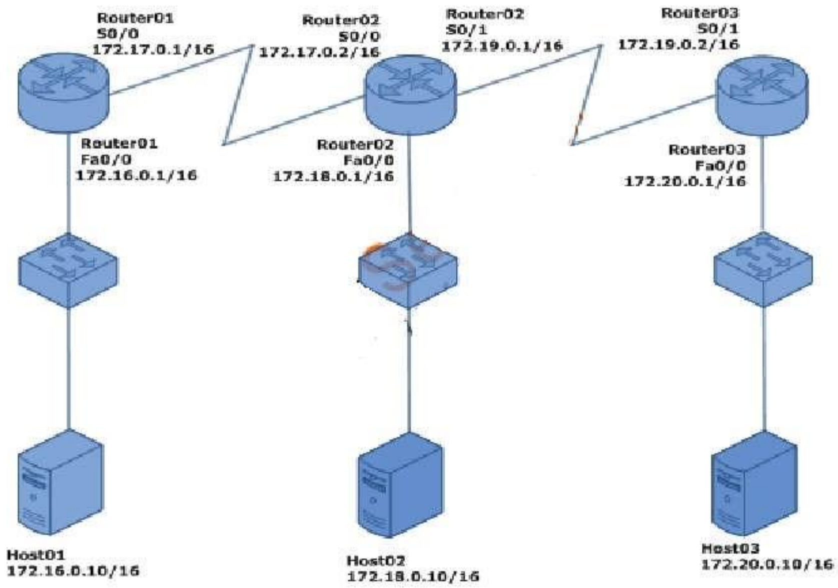
Static Routing

Without configuration of static or dynamic routes you cannot transmit packets outside the network.

Static routing has the following steps:-

- Configure host names and interfaces i.e assign IP addresses to all interfaces (in use) of all routers
- Configure static routes.

Create the following topology and assign IP-addresses to the interfaces as mentioned in this figure.



Step 1: Hostname, interface, and IP address configurations in all routers.

➢ Configurations in Router01

Connect to Router01 console and use the following IOS commands to configure host name as Router01.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router01
Router01(config)#

For Router interface configuration type "interface interface_name".

Router01(config)#interface fa0/0 router01(config-if)#

Assign an IP address to an interface. Run the following command from interface configuration mode.

Router01(config-if)#ip address 172.16.0.1 255.255.0.0

You have to enter both IP address and subnet mask.

Now enable router01 interface. Run "no shutdown" command from interface configuration mode.

Router01(config-if)#no shutdown
Router02(config-if)#exit
LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Note: To disable a router interface we run "shutdown" command from interface configuration mode.

Use the following IOS commands to open the serial interface S2/0 configuration mode on Router01 and configure IP address as 172.17.0.1/16. You have to set a clock rate also using the "clock rate" command on S0/0 interface, since this is the DCE side.

```
Router01(config)#interface se2/0
Router01(config-if)#clock rate 64000
Router01(config-if)#ip address 172.17.0.1  255.255.0.0
Router01(config-if)#no shutdown
Router02(config-if)#exit
```

Do remember to run the "copy running-config startup-config" command from enable mode, if you want to save the changes you have made in the router.

➢ Configuration in Router02

Connect to Router02 console and use the following IOS commands to configure host name as Router02.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router02

Router02(config)#interface fa0/0
Router02(config-if)#ip address 172.18.0.1 255.255.0.0
Router02(config-if)#no shutdown
Router02(config-if)#exit

Router02(config)#interface s2/0
Router02(config-if)#ip address 172.17.0.2 255.255.0.0
Router02(config-if)#no shutdown
Router02(config-if)#exit
```

Use the following IOS commands to open the serial interface S3/0 configuration mode on Router02 and configure IP address as 172.19.0.1/16. You have to set a clock rate also using the "clock rate" command on S0/1 interface, since this is the DCE side.

```
Router02(config)#interface s3/0
Router02(config-if)#clock rate 64000
Router02(config-if)#ip address 172.19.0.1 255.255.0.0
Router02(config-if)#no shutdown
Router02(config-if)#exit
```

Do remember to run the "copy running-config startup-config" command from enable mode, if you want to save the changes you have made in the router.

➢ Configurations in Router03

Connect to Router03 console and use the following IOS commands to configure host name as Router03.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router03

Router03(config)#interface fa0/0
Router03(config-if)#ip address 172.20.0.1   255.255.0.0
Router03(config-if)#no shutdown
Router02(config-if)#exit

Router03(config)#interface s2/0
Router03(config-if)#ip address 172.19.0.2   255.255.0.0
Router03(config-if)#no shutdown

Do remember to run the "copy running-config startup-config" command from enable mode, if you want to save the changes you have made in the router.. **Step 2: Configure Static Routes**

Static Route can be configured by the following IOS commands.

- Router(config)#ip route   destination_network   subnet_mask   default_gateway

OR

- Router(config)# ip route destination_network subnet_mask interface_to_exit [administrative_distance] [permanent]

The permanent keyword will keep the static route in the routing table even when the interface the router uses for the static route fails.

- **Static Routing configuration in Router01**

Connect to Router01 console and use the following IOS commands to configure static routing in Router01. The "ip route" commands shown below states that to reach 172.18.0.0/16, 172.19.0.0/16 and 172.20.0.0/16 networks, handover the packets to the gateway ip address 172.17.0.2. The networks 172.16.0.0/16 and 172.17.0.0/16 are conneted directly to Router01.

Router01>enable
Router01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z. Router01(config)#ip route
172.18.0.0   255.255.0.0   172.17.0.2
Router01(config)#ip route 172.19.0.0   255.255.0.0   172.17.0.2
Router01(config)#ip route 172.20.0.0   255.255.0.0   172.17.0.2

Do remember to run the "copy running-config startup-config" command from enable mode, if you want to save the changes you have made in the router.

To view the routing table in Router01, run "show ip route" command in Router01 as shown below.

```
Router01#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP
external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external
type 1, E2 - OSPF external type 2, E - EGP i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia
- IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
C    172.16.0.0/16 is directly connected, FastEthernet0/0
C    172.17.0.0/16 is directly connected, Serial0/0
S    172.18.0.0/16 [1/0] via 172.17.0.2
S    172.19.0.0/16 [1/0] via 172.17.0.2
S    172.20.0.0/16 [1/0] via 172.17.0.2
```

The "S" character at the beginning of a line in routing table shows that it is a static route and "C" character shows that it is a directly connected network.

- **Static Routing configuration in Router02**

Connect to Router02 console and use the following IOS commands to configure static routing in Router02. The "ip route" commands shown below states that to reach 172.16.0.0/16 network, handover the packets to the gateway ip address 172.17.0.1 and to reach 172.20.0.0/16 network, handover the packets to the gateway ip address 172.19.0.2. The networks 172.17.0.0/16, 172.18.0.0/16 and 172.19.0.0/16 are connected directly to Router02.

```
Router02>enable
Router02#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router02(config)#ip route 172.16.0.0 255.255.0.0 172.17.0.1
Router02(config)#ip route 172.20.0.0 255.255.0.0 172.19.0.2
```

Do remember to run the "copy running-config startup-config" command from enable mode, if you want to save the changes you have made in the router.

To view the routing table in Router02, run "show ip route" command in Router02 as shown below.

```
Router02#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external
type 1, E2 - OSPF external type 2, E - EGP i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia
- IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
S    172.16.0.0/16 [1/0] via 172.17.0.1
C    172.17.0.0/16 is directly connected, Serial0/0
C    172.18.0.0/16 is directly connected, FastEthernet0/0
C    172.19.0.0/16 is directly connected, Serial0/1
```

S 172.20.0.0/16 [1/0] via 172.19.0.2

The "S" character at the beginning of a line in routing table shows that it is a static route and "C" character shows that it is a directly connected network.

- **Static Routing configuration in Router03**

Connect to Router03 console and use the following IOS commands to configure static routing in Router03. The "ip route" commands shown below states that to reach 172.16.0.0/16, 172.17.0.0/16 and 172.18.0.0/16 networks, handover the packets to the gateway ip address 172.19.0.1. The networks 172.19.0.0/16 and 172.20.0.0/16 are conneted directly to Router03.

```
Router03>enable
Router03#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router03(config)#ip route 172.16.0.0 255.255.0.0 172.19.0.1
Router03(config)#ip route 172.17.0.0 255.255.0.0 172.19.0.1
Router03(config)#ip route 172.18.0.0 255.255.0.0 172.19.0.1
```

Do remember to run the "copy running-config startup-config" command from enable mode, if you want to save the changes you have made in the router.

To view the routing table in Router03, run "show ip route" command in Router03 as shown below.

```
Router03#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external
type 1, E2 - OSPF external type 2, E - EGP i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia
- IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
S    172.16.0.0/16 [1/0] via 172.19.0.1
S    172.17.0.0/16 [1/0] via 172.19.0.1
S    172.18.0.0/16 [1/0] via 172.19.0.1
C    172.19.0.0/16 is directly connected, Serial0/1
C    172.20.0.0/16 is directly connected, FastEthernet
```

The "S" character at the beginning of a line in routing table shows that it is a static route and "C" character shows that it is a directly connected network.

Verify the connectivity between networks using the ping command

To verify the static routes which we have configured and the connectivity between networks, run the ping command from Host01 (IP address: 172.16.0.10/16) to Host03 (IP address: 172.20.0.10/16).

C:\>ping 172.20.0.10

Pinging 172.20.0.10 with 32 bytes of data:

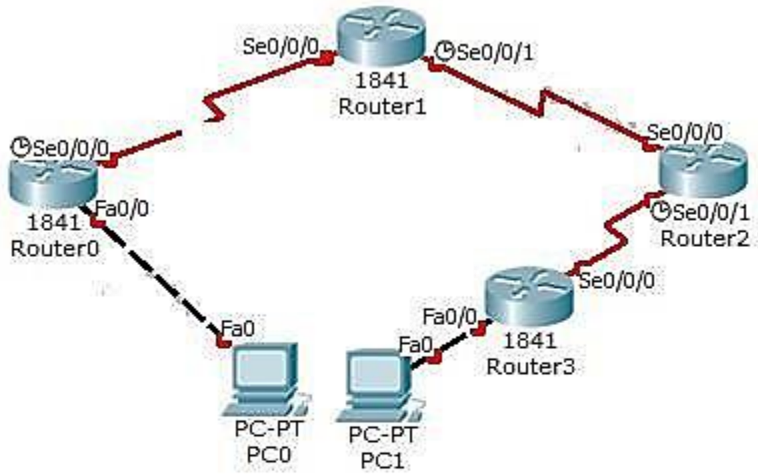
Reply from 172.20.0.10: bytes=32 time=172ms TTL=125
Reply from 172.20.0.10: bytes=32 time=235ms TTL=125
Reply from 172.20.0.10: bytes=32 time=187ms TTL=125
Reply from 172.20.0.10: bytes=32 time=187ms TTL=125

Ping statistics for 172.20.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times
in milli-seconds:
Minimum = 172ms, Maximum = 235ms, Average = 195ms

The ping reply from Host03 (IP address: 172.20.0.10/16) shows that the static routes are configured well in three routers and there is network connectivity between different networks.

LAB TASK:

Create below mentioned topology and implement Static routing:



IP Configuration:

Device	Connected from	Connected to	IP Address
PC0	FastEthernet0	Router0's FastEthernet0/0	10.0.0.2/8
Router0	FastEthernet0/0	PC0's FastEthernet0	10.0.0.1/8
Router0	Serial0/0/0	Router1's serial0/0/0	192.168.0.253/30
Router1	Serial0/0/0	Router0's Serial0/0/0	192.168.0.254/30
Router1	Serial0/0/1	Router2's Serial0/0/0	192.168.0.249/30
Router2	Serial0/0/0	Router1's Serial0/0/1	192.168.0.250/30
Router2	Serial0/0/1	Router3's Serial0/0/0	192.168.0.245/30
Router3	Serial0/0/0	Router2's Serial0/0/1	192.168.0.246/30
Router3	FastEthernet0/0	PC1's FastEthernet0	20.0.0.1/8
PC1	FastEthernet0	Router1's FastEthernet0/0	20.0.0.2/8

1. Show static route of each router. How will you find out static route.
2. Is ping successful from PC0 **<->** PC1? Show ping results.